```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
df = pd.read_excel('/content/Sample data (1).xlsx')

df.head()
```

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	D
0	Government	Germany	Carretera	NaN	1513.0	3	350	529550.0	0.0	529550.0	393380.0	136170.0	20 12
1	Government	Germany	Paseo	NaN	1006.0	10	350	352100.0	0.0	352100.0	261560.0	90540.0	20 06
2	Government	Canada	Paseo	NaN	1725.0	10	350	603750.0	0.0	603750.0	448500.0	155250.0	20 11
3	Government	Germany	Paseo	NaN	1513.0	10	350	529550.0	0.0	529550.0	393380.0	136170.0	20 12
4	Government	Germany	Velo	NaN	1006.0	120	350	352100.0	0.0	352100.0	261560.0	90540.0	20 06

Next steps: Generate code with df | New interactive sheet

```
print("Dataset Shape:")
print(df.shape)

print("Column Names:")
print(df.columns)

print("Dataset Information:")
df.info()
```

```
Dataset Shape:
(700, 16)
Column Names:
Index(['Segment', 'Country', 'Product', 'Discount Band', 'Units Sold',
       'Manufacturing Price', 'Sale Price', 'Gross Sales', 'Discounts',
       ' Sales', 'COGS', 'Profit', 'Date', 'Month Number', 'Month Name',
       'Year'],
      dtype='object')
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 700 entries, 0 to 699
Data columns (total 16 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Segment              700 non-null    object
 1   Country              700 non-null    object
 2   Product              700 non-null    object
 3   Discount Band        647 non-null    object
 4   Units Sold           700 non-null    float64
 5   Manufacturing Price  700 non-null    int64
 6   Sale Price           700 non-null    int64
 7   Gross Sales          700 non-null    float64
 8   Discounts            700 non-null    float64
 9    Sales               700 non-null    float64
 10  COGS                 700 non-null    float64
 11  Profit               700 non-null    float64
 12  Date                 700 non-null    datetime64[ns]
 13  Month Number         700 non-null    int64
 14  Month Name           700 non-null    object
 15  Year                 700 non-null    int64
dtypes: datetime64[ns](1), float64(6), int64(4), object(5)
memory usage: 87.6+ KB
```

```
print(df.isnull().sum())
```

```
Segment                0
Country                0
Product                0
Discount Band         53
Units Sold             0
```

```
Manufacturing Price     0
Sale Price              0
Gross Sales             0
Discounts               0
 Sales                  0
COGS                    0
Profit                  0
Date                    0
Month Number            0
Month Name              0
Year                    0
dtype: int64
```

```
print("Duplicate Rows:")
print(df.duplicated().sum())
```

```
Duplicate Rows:
0
```

```
# Remove extra spaces from column names
df.columns = df.columns.str.strip()

# Fill missing discount bands
df['Discount Band'] = df['Discount Band'].fillna('No Discount')

# Remove duplicate records
df = df.drop_duplicates()

# Convert date column
df['Date'] = pd.to_datetime(df['Date'])

# Check missing values again
print(df.isnull().sum())

print("Data cleaning completed!")
```

```
Segment               0
Country               0
Product               0
Discount Band         0
Units Sold            0
Manufacturing Price   0
Sale Price            0
Gross Sales           0
Discounts             0
Sales                 0
COGS                  0
Profit                0
Date                  0
Month Number          0
Month Name            0
Year                  0
dtype: int64
Data cleaning completed!
```

```
df.to_excel(
    'Cleaned_Sales_Data.xlsx',
    index=False
)
```

```
from google.colab import files

files.download('Cleaned_Sales_Data.xlsx')
```

```
total_sales = df['Sales'].sum()
total_profit = df['Profit'].sum()
total_units = df['Units Sold'].sum()

print("Total Sales:", total_sales)
print("Total Profit:", total_profit)
print("Total Units Sold:", total_units)
```

```
Total Sales: 118726350.25999999
Total Profit: 16893702.26
Total Units Sold: 1125806.0
```

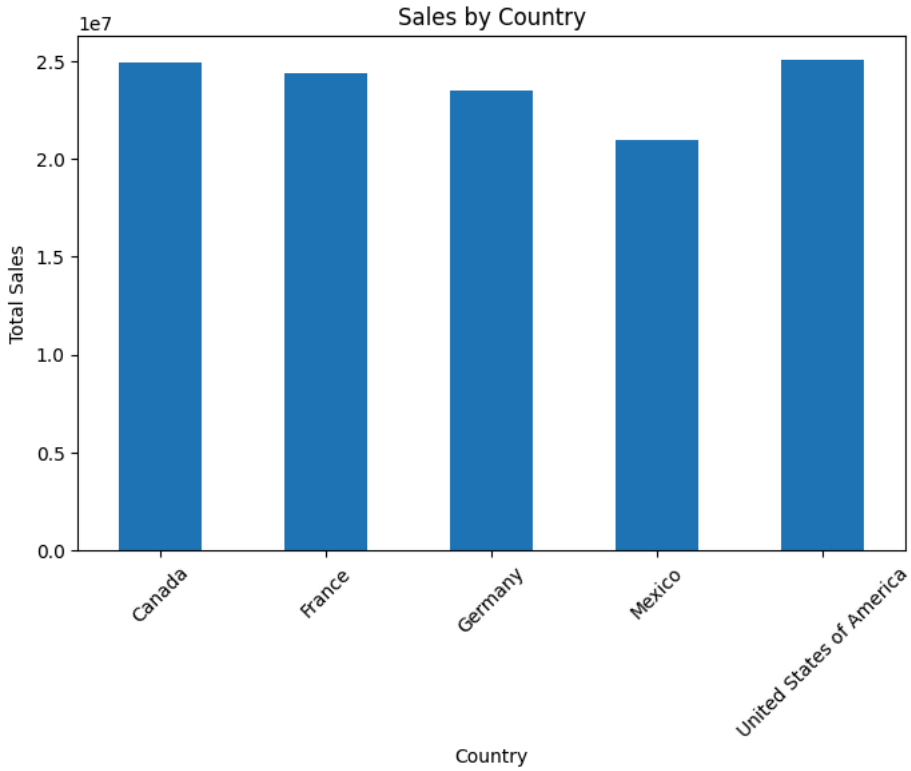
```
country_sales = df.groupby('Country')['Sales'].sum()

print(country_sales)
```

```
Country
Canada                        2.488765e+07
France                        2.435417e+07
Germany                       2.350534e+07
Mexico                        2.094935e+07
United States of America      2.502983e+07
Name: Sales, dtype: float64
```

```
country_sales.plot(
    kind='bar',
    figsize=(8,5)
)

plt.title('Sales by Country')
plt.xlabel('Country')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.show()
```



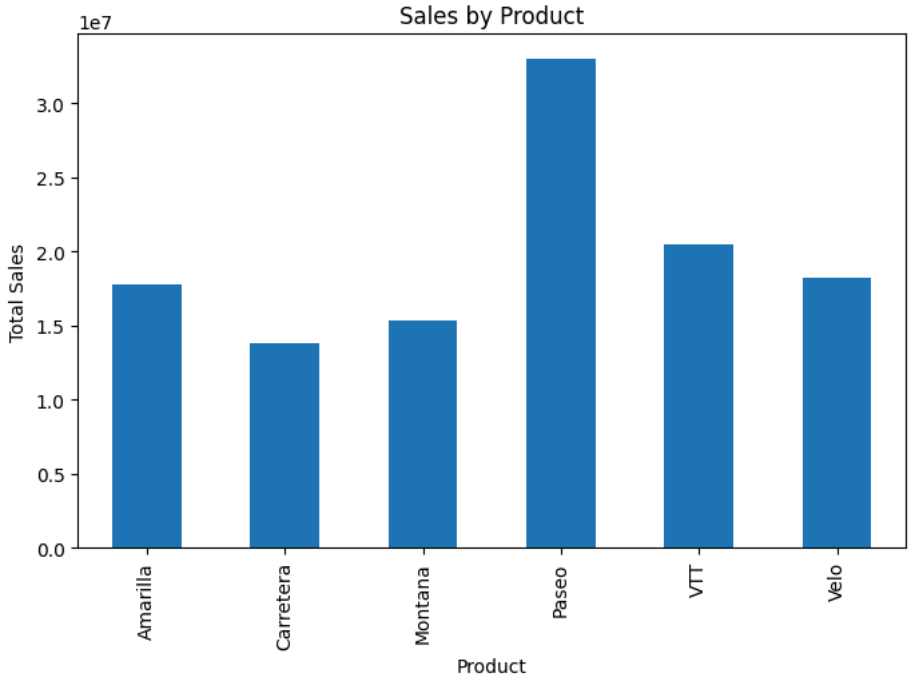

```
product_sales = df.groupby('Product')['Sales'].sum()

print(product_sales)
```

```
Product
Amarilla     1.774712e+07
Carretera    1.381531e+07
Montana      1.539080e+07
Paseo        3.301114e+07
VTT          2.051192e+07
Velo         1.825006e+07
Name: Sales, dtype: float64
```

```
product_sales.plot(
    kind='bar',
    figsize=(8,5)
)

plt.title('Sales by Product')
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.show()
```



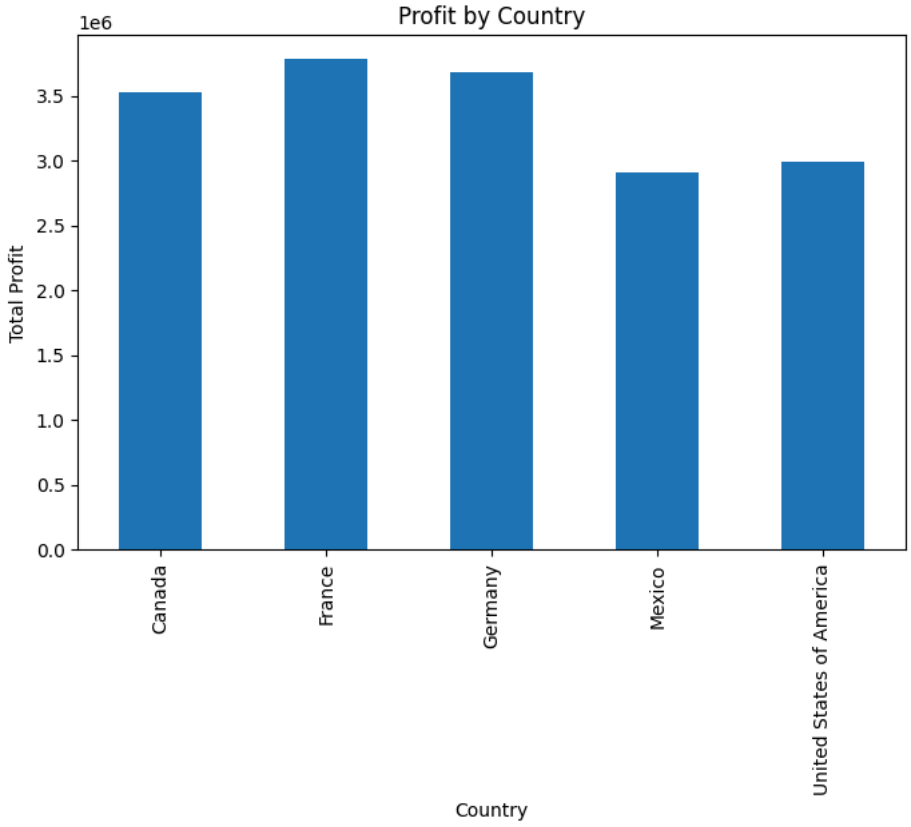

```
profit_country = df.groupby('Country')['Profit'].sum()

print(profit_country)

profit_country.plot(kind='bar', figsize=(8,5))

plt.title('Profit by Country')
plt.xlabel('Country')
plt.ylabel('Total Profit')
plt.show()
```

```
Country
Canada                      3529228.885
France                      3781020.780
Germany                     3680388.820
Mexico                      2907523.110
United States of America    2995540.665
Name: Profit, dtype: float64
```



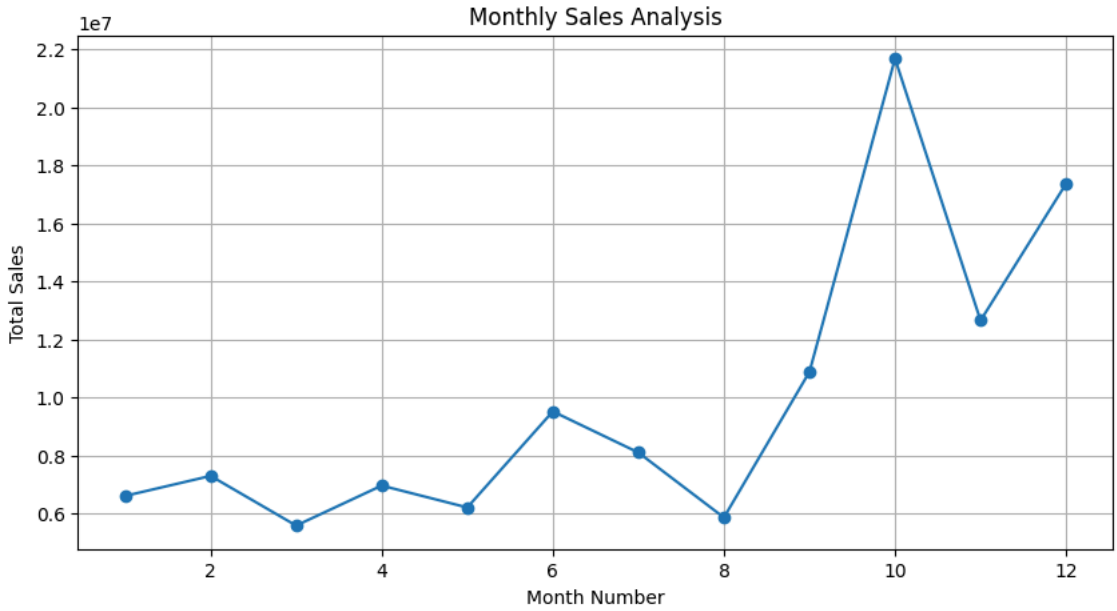
```
monthly_sales = df.groupby(
    'Month Number'
)['Sales'].sum()

print(monthly_sales)

monthly_sales.plot(
    kind='line',
    marker='o',
    figsize=(10,5)
)

plt.title('Monthly Sales Analysis')
plt.xlabel('Month Number')
plt.ylabel('Total Sales')
plt.grid(True)
plt.show()
```

```
Month Number
1      6607761.68
2      7297531.39
3      5586859.87
4      6964775.07
5      6210211.06
6      9518893.82
7      8102920.18
8      5864622.42
9     10882697.27
10    21671431.02
11    12651417.50
12    17367228.98
Name: Sales, dtype: float64
```

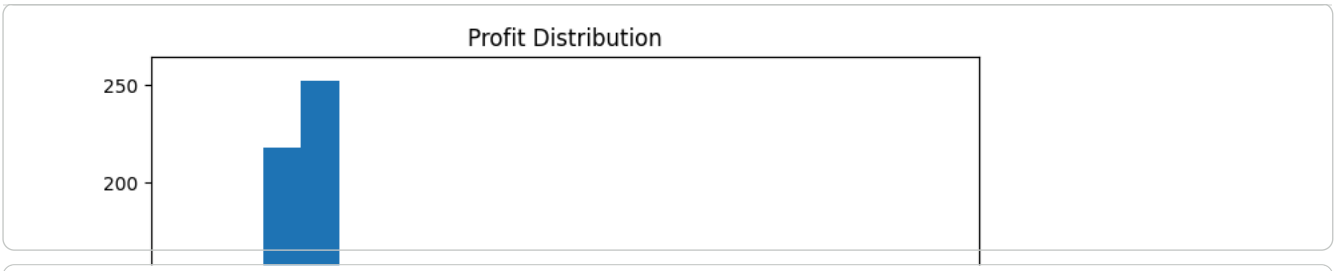



```
plt.figure(figsize=(8,5))

plt.hist(df['Profit'], bins=20)

plt.title('Profit Distribution')
plt.xlabel('Profit')
plt.ylabel('Frequency')

plt.show()
```




```
plt.figure(figsize=(10,6))

sns.heatmap(
    df.select_dtypes(include='number').corr(),
    annot=True,
    cmap='Blues'
)

plt.title('Correlation Heatmap')
plt.show()
```

